# FastLane: A Voice Agent That Routes Each Turn Between a Small and a Large Model

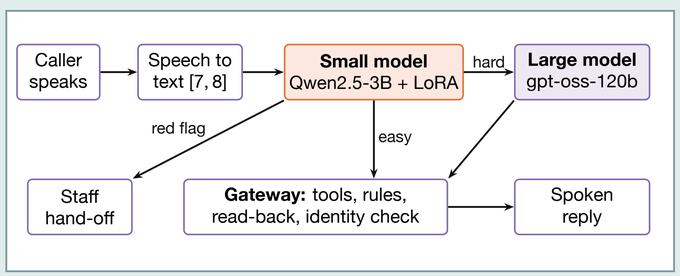
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## ABSTRACT

Most voice agents send every turn to a large API model, which is slow and costly, although most turns of a booking call are easy [1, 2]. FastLane books, changes and cancels movie tickets, bus tickets and restaurant tables for Indian-English callers. A fine-tuned Qwen2.5-3B [3] answers every turn first and passes only hard turns to gpt-oss-120b [4] or to staff, while a gateway enforces the booking rules and a read-back confirmation in code. On 120 simulated calls FastLane succeeds in 90% (large model alone: 82%) with 78% fewer large-model calls and no booking change the caller did not confirm. It is not faster on our hardware.

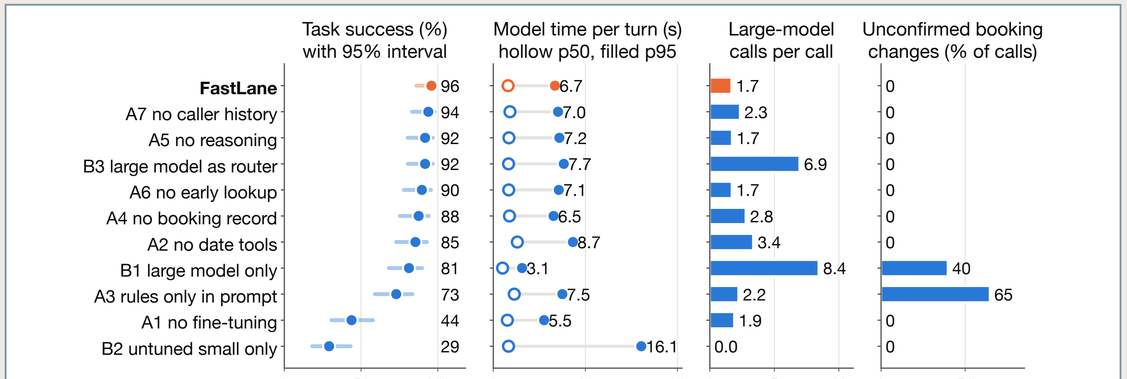
## METHODOLOGY

- **Practice world:** SQLite booking system for three domains, 11 tools, 360 scripted test calls.
- **Safety in code:** rules, read-back before every change and identity checks live in a gateway, not in the prompt.
- **Training data without an LLM teacher:** rule-based callers and an oracle agent run 650 practice calls through the real agent; QLoRA [5] on a Kaggle T4, served as 4-bit MLX on a laptop.
- **Evaluation:** 12 systems, a simulated caller (Qwen2.5-7B) as in $\tau$-bench [6], paired McNemar tests, and every failed call read by hand.



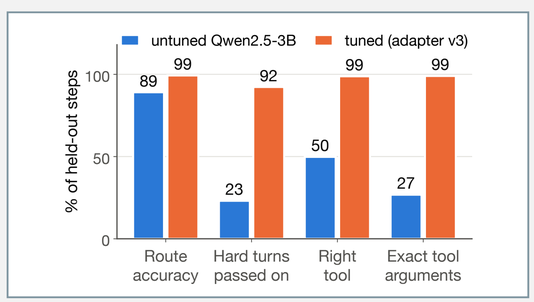

## RESULTS

Same 48 test calls for every system (FastLane and B1 also on 120), one seed, gpt-oss-120b on Groq. With rules in code FastLane made **no unconfirmed booking change**, against 44 of 120 calls for B1 (large model only) and 31 of 48 for A3 (rules only in the prompt). It succeeded at least as often as B1 (90.0% vs. 81.7%, $p = 0.087$) with 78% fewer large-model calls, but B1 is faster (p95 3.0 s vs. 6.8 s per turn).



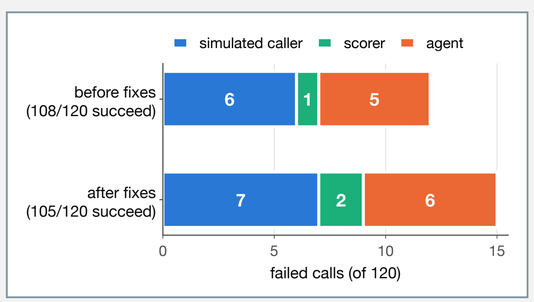

## FINE-TUNING MATTERS

On 267 held-out steps the untuned 3B model fills tool arguments exactly in 27% of steps, the tuned one in 99%. End to end, the untuned model succeeds in 29% of calls alone (B2) and in 44% inside FastLane (A1), against 96%. A model never trained on restaurants still gets 37 of 40 restaurant calls right.




## READING EVERY FAILURE

Half of FastLane's 12 failures came from the simulated caller, mostly by not changing its mind as scripted. The 5 agent errors (a guessed bus origin, a false "cancelled", a contradicted policy text) led to four checks in code. They removed those errors, but success in a rerun moved only within noise (13 of 120 calls changed, $p = 0.58$).




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## CONCLUSION AND FUTURE WORKS

- A tuned 3B model handles four of five turns, with 78% fewer large-model calls.
- Safety belongs in code: no unconfirmed changes, vs. 37–65% of calls with rules in the prompt.
- No speed gain here: a hosted 120B model beats a 3B model on a laptop.
- Next: a faster small-model path, retraining on corrected data, real callers.

## ACKNOWLEDGMENT

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## REFERENCES

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**Code:** github.com/Sankalp0109/FastLane-Adaptive-Voice-Agent

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